Unlocking Flexibility Potential in Unit Commitment through DVFS - driven Data Center Clusters

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***Abstract*—This document describes the most common article elements and how to use the IEEEtran class with LATEX to produce files that are suitable for submission to the Institute of Electrical and Electronics Engineers (IEEE). IEEEtran can produce conference, journal and technical note (correspondence) papers with a suitable choice of class options.**

***Index Terms*—Class, IEEEtran, LATEX, paper, style, template, typesetting.**

I. Introduction

As the increase in various artificial intelligence technologies and demand energy consumptions, the operation of electrical power systems are becoming diverse and uncontrollable, and they are facing challenges in provide sufficient flexible adjustment resources against emergence events. In such circumstances, additional flexible mechanisms such as demand response (DR), virtual Power plant (VPP), and energy storage systems (ESS), etc., needs to be prioritized. Nevertheless, the data center clusters (DCCs), which act as a significant energy consumer, have not been fully utilized in the flexible adjustment resources analysis. To this end, it it essential to investigate the flexible adjustment characteristic of DCCs, and to explore the potential of DCCs in providing flexible adjustment resources for the economic and reliable operation of electrical power systems.

Recent literature reported several studies on the DCCs flexibility modeling and its application in electrical power systems operation. Specially, Ref. [1] carried out a Quality-of-service-based cost function to describe the workload shifting and server utilization of DCCs, and extended a distributed dispatch framework with considering DCCs flexibility. Ref. [2] designed a mix-integer programming model to achieve the the optimal demand management of DCCs. Ref. [3] analyzed the progressive load characteristics in DCCs, and proposed two optimization models to estimate the optimal capability planning of storage systems in DCCs and design parameters optimizations. Ref. [4] explored the statistical feasibility of DDCs and hydrogen resources in a microgrid context, and carried out a data-driven statistical-feasibility-based robust rolling optimization framework to analysis the optimal operation of DDCs and hydrogen storage systems under complex uncertain factors. Ref. [5] further analyzed the potential capability of DDCs for regulating computational workloads while reducing carbon emissions, and then constructed a emission-aware coordinated dispatch method of multi-regional microgrids with DDCs. The aforementioned studies primarily emphasize the flexibility ability of DCCs within electrical power system operations. However, they often overlook the intricate physical characteristics of DCCs, particularly in relation to workload shifting dynamics and the implementation of advanced high-evolution control strategies.

The workloads in DCCs can be broadly classified into three categories: i) $compute\text{-}intensive\ workloads$, which include high-performance computing (HPC), machine learning model training, and commercial application services. ii) $storage\text{-}intensive\ workloads$, which are employed for tasks such as regular file backups, archival systems, and relational database applications. iii) $network\text{-}intensive\ workloads$, encompassing activities like website hosting, content delivery networks, and network traffic management. Notably, compute-intensive and storage-intensive workloads dominate in DCCs, collectively accounting for approximately 70% of the total workload. This dominance is driven by the growing demand for HPC and data-driven techniques. In contrast, network-intensive workloads are typically less predictable and highly time-sensitive, requiring immediate processing to ensure seamless operation. Since the former two workloads are more predictable and flexible, the peak workload requirements of the former two workloads can be effectively shifted or adjusted through dynamic voltage frequency scaling (DVFS) to achieve load balancing and energy management.

Driven by such motivation, this paper constructs a group of DVFS-driven physical constraints for representing the flexible characteristic of various workload configurations, and further extends an enhanced unit commitment (EUC) model to reflect actively response of DCCs under complex uncertain environment. The main contributions of this paper are summarized as follows:

- A DVFS-driven physical constraints is developed to characterize the flexibility attributes of DCCs workloads. Furthermore, the proposed constraints are linearized using McCormick Envelopes to enhance computational tractability in UC studies.
- A EUC model is formulated to capture the dynamic flexible response of DCCs. The model incorporates a new coordination mechanism to address multi-timescale interactions between workload shifting in DCCs and grid-side demand fluctuations.

The remainder of this abstract is organized as follows. Section II outlines the scheduling framework for power systems, incorporating the consumption characteristics of DCC workloads. Section III proposed DVFS-driven physical constraints.

II. The proposed Scheduling Framework

This section presents a new hierarchical scheduling framework designed to integrate the flexible response capabilities of DCCs. Recognizing the difference in scheduling timescales (hourly for grid-side operations versus seconds for DCC workload management), the framework is structured into two interconnected layers. The upper layer encompasses the grid-side scheduling model, while the lower layer represents the DCC workload management model. The upper layer optimizes the power system's generation schedule, and the lower layer optimizes workload shifting within DCCs. A coordination mechanism links these layers, facilitating seamless communication and synchronization to achieve a cohesive and efficient scheduling process.

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